Code-Layout, Readability and Reusability

Date	21 October 2026
Team ID	VAI-AI-2026
Project Name	VoyageAI — Personalized AI Travel Itinerary Planner
Maximum Marks	5 Marks

Code Quality & Clean Architecture Standards:

The codebase adheres to clean code conventions, SOLID principles, type hinting, and modular separation.

Directory / Module	Design Pattern / Standard	Reusability & Maintainability Benefit
backend/app/ai/	Strategy Pattern & Client Wrapper	Decoupled LLM provider; easily switch or fallback between models without changing router logic.
backend/app/models/	SQLAlchemy 2.0 Declarative Base	Unified relational ORM with cascade deletes and strict foreign-key integrity.
frontend/src/store/	Zustand Lightweight Store	Centralized reactive state for auth and trips without prop-drilling or Redux boilerplate.
frontend/src/components/	Atomic Component Design	Reusable UI widgets (Navbar, ActivityModal, TimelineCard, Badge) across multiple views.